

IBM University Engagement Project Submission

“Exploring the Power of Agentic AI with IBM Granite and LangFlow”

Project Tile – Recipe Generator Agent

Domain of Project – Food & Recipe

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Problem Statement - Document Q&A RAG Agent for Recipe Generator

The Challenge

Food enthusiasts and culinary professionals often rely on unstructured recipe documents, cookbooks, blogs, and user-submitted notes. Searching across these scattered resources is time consuming, and it's difficult to adapt recipes based on dietary needs, available ingredients, or cultural preferences. The challenge is not just to retrieve recipes but to intelligently query, adapt, and generate personalized cooking instructions.

The Objective

Build a Document Q&A RAG (Retrieval-Augmented Generation) Agent that can act as an intelligent recipe generator.

The solution should:

1. Ingest & Index Recipe Documents: Parse cookbooks, PDFs, blogs, or uploaded recipe notes and store them in a searchable knowledge base.
2. Enable Conversational Q&A: Allow users to ask natural language questions (e.g., "How can I make a sugar-free version of chocolate cake?") and receive precise, recipe-aware answers.
3. Create a Personalized & Agentic Layer: Adapt recipes based on user constraints such as available ingredients, dietary restrictions, cooking time, or cuisine preferences.
4. Visualize Cooking Guidance: Present results as step-by-step cooking instructions with substitution suggestions, nutritional facts, and even generate a shopping list.

Proposed Solution -

AI-Powered Recipe Generator Agent

The proposed solution is an **AI-powered Document Q&A RAG Agent** built using **LangFlow**, **IBM watsonx.ai Granite Models**, and a **Vector Knowledge Base**. It enables users to upload recipe documents such as PDFs, cookbooks, blogs, and personal notes, making them searchable through natural language conversations.

The system follows a **Retrieval-Augmented Generation (RAG)** workflow. Recipe documents are parsed, split into chunks, embedded, and stored in a vector database. When a user asks a question, the system retrieves the most relevant recipe information and uses Granite Models to generate accurate, context-aware responses.

The agent can answer cooking-related questions, recommend ingredient substitutions, adapt recipes based on dietary preferences or available ingredients, generate step-by-step cooking instructions, and provide nutritional insights and shopping list suggestions.

LangFlow orchestrates the complete workflow, **IBM watsonx.ai Granite Models** power intelligent reasoning and recipe generation, and the **vector knowledge base** ensures fast and accurate semantic retrieval. Overall, the solution transforms unstructured recipe documents into an intelligent, personalized cooking assistant.

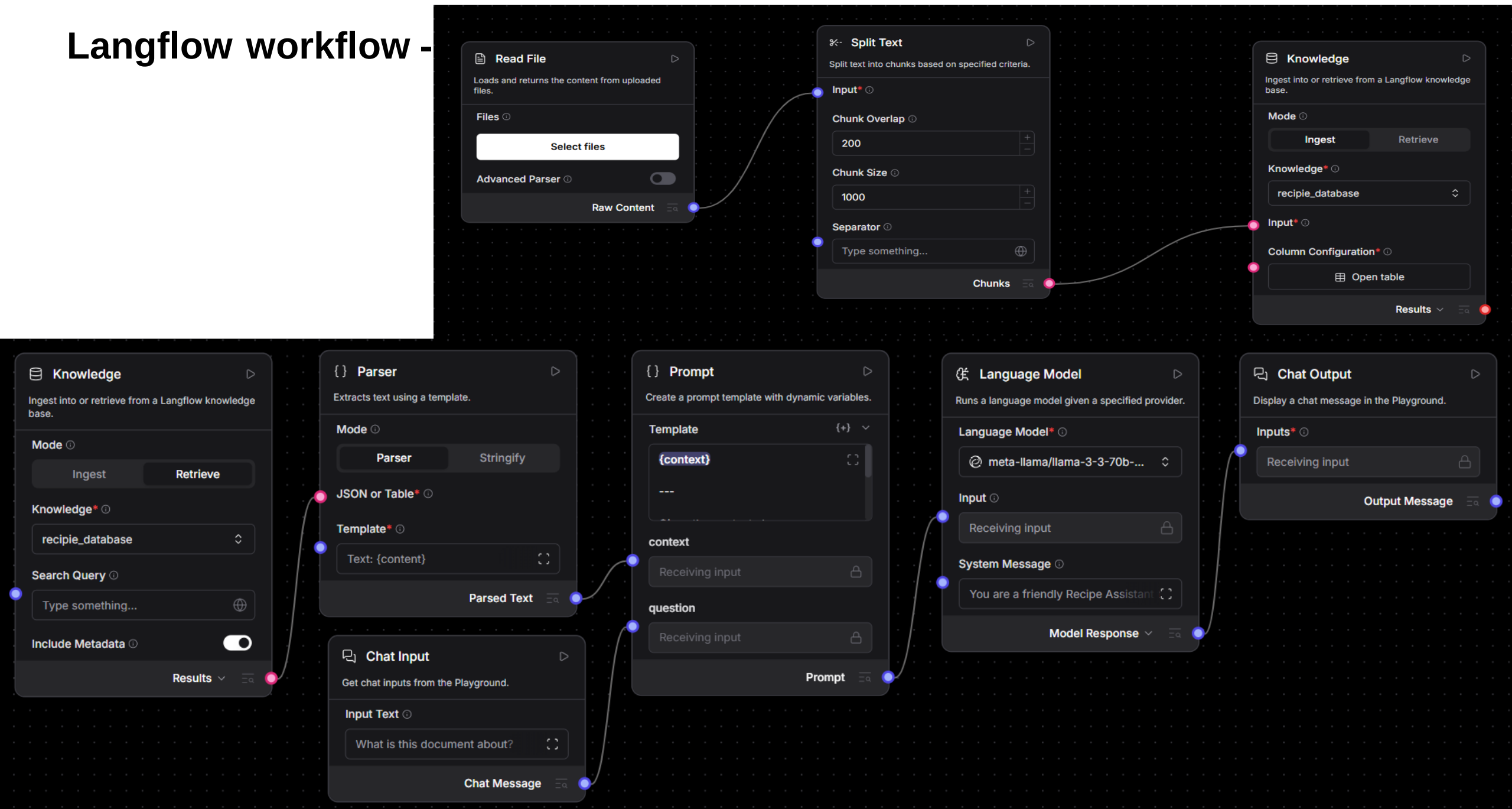
Technology Used

- **Langflow platform** -For designing and orchestrating multi-agent workflows visually
- **IBM Granite model** - for eg – ibm-granite-3-2-8b – For natural language understanding, reasoning, and personalization
- **IBM Cloud** – It provides the infrastructure to build, deploy, and scale the Nutrition Agent efficiently. IBM Cloud enables a **reliable, secure, and scalable environment** for deploying the Nutrition Agent.
- **RAG**– RAG to **fetch real data first, then generate intelligent answers**, making the Nutrition Agent more trustworthy and effective.
- **IBM Watsonx.ai** - For model deployment, embeddings, and AI governance
- **Vector Database** (FAISS/Chroma) – For RAG-based nutritional data retrieval

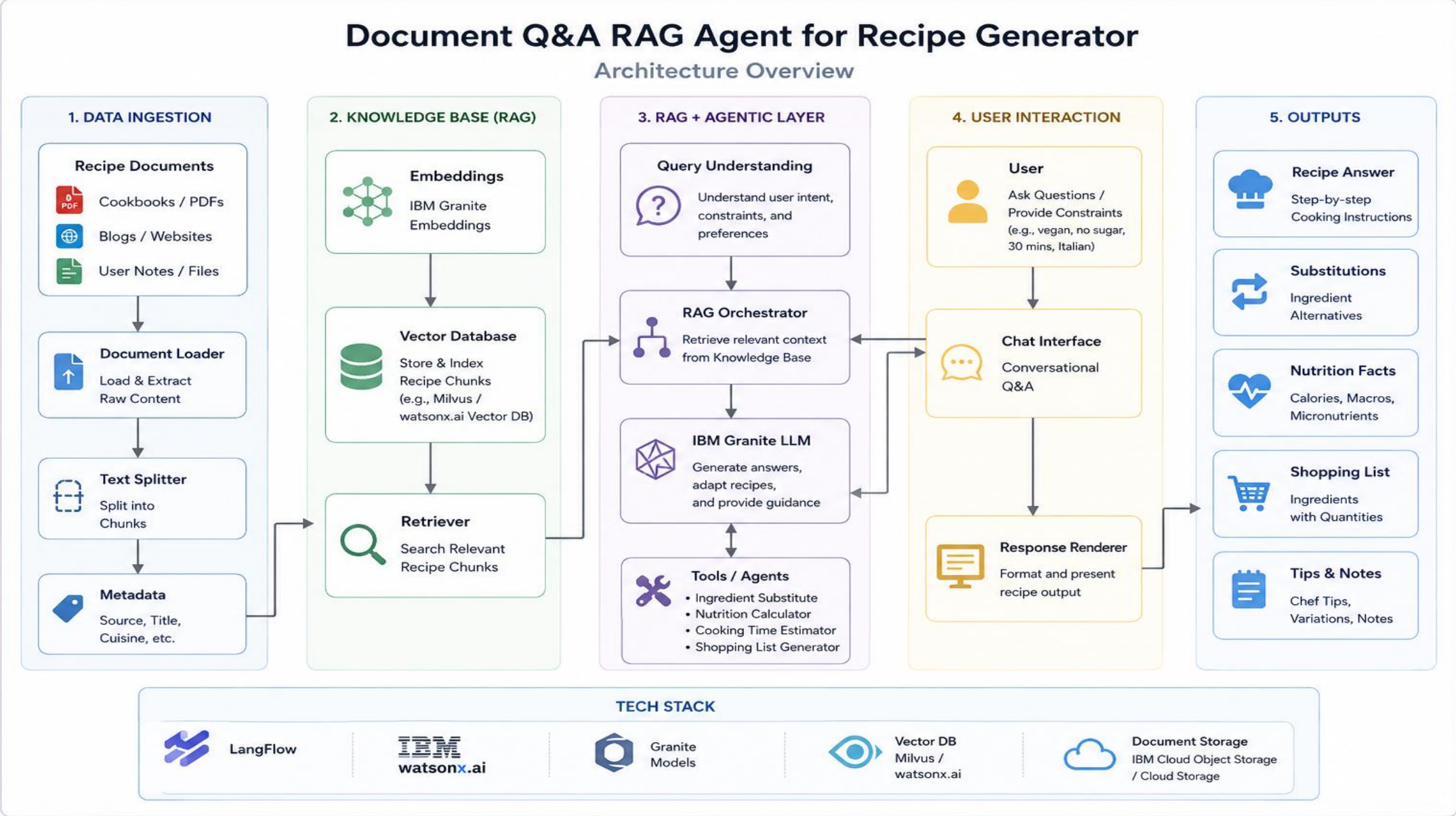
Langflow component Used -

- 1) **Chat Input** – Interface where users enter their queries, preferences, or meal details (text/voice/image).
- 2) **IBM watsonx AI Agent** - Processes user input using Granite models to generate personalized nutrition insights and recommendations.
- 3) **Name of IBM Granite model** - Granite-4.0-8B-Instruct → Fast, efficient for real-time diet recommendations
- 4) **Chat output** -Displays AI-generated responses such as diet plans, nutritional analysis, and health suggestions.
- 5) **File component** - Stores and retrieves user data, meal logs, and nutritional datasets for analysis and tracking.

Langflow workflow -



Architecture blueprint



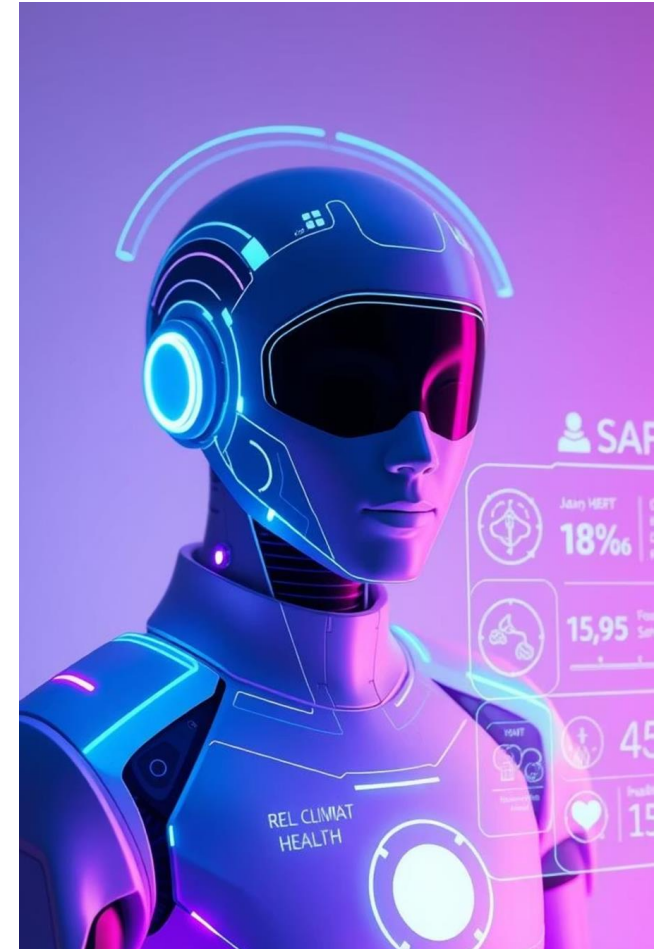
Role of Agentic AI in the solution

Role of Agentic AI in Recipe Generator Agent

Agentic AI enables the Recipe Generator to function as an intelligent assistant rather than a simple search system. Using **IBM watsonx.ai** and **Granite Models**, it understands user queries, retrieves the most relevant recipe information from the knowledge base, and generates personalized, context-aware responses.

The agent adapts recipes based on user preferences such as dietary restrictions, available ingredients, cooking time, and cuisine choices. It can also suggest ingredient substitutions, provide step-by-step cooking instructions, and generate shopping lists and nutritional insights.

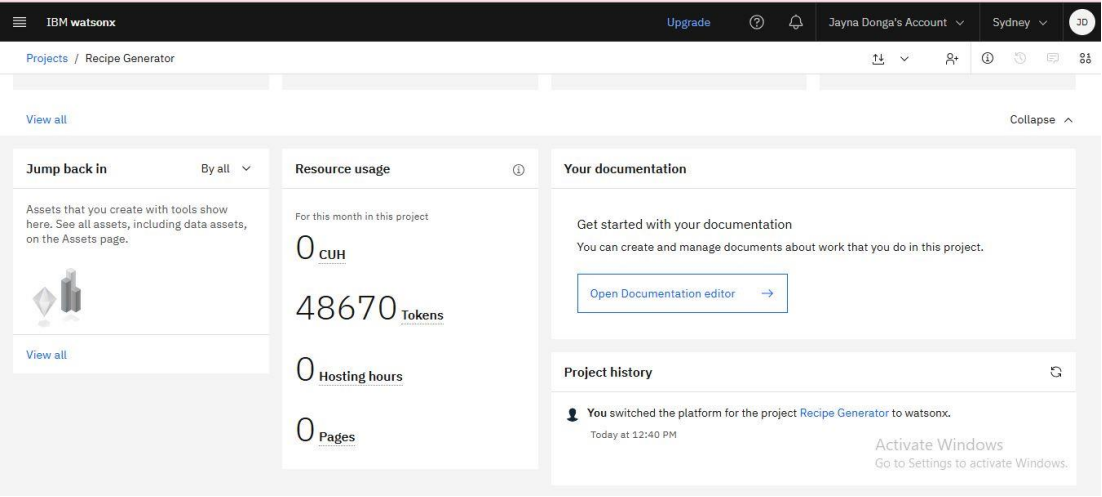
By combining **RAG** with intelligent reasoning, Agentic AI delivers accurate, personalized, and conversational cooking assistance, making recipe discovery faster and more user-friendly.



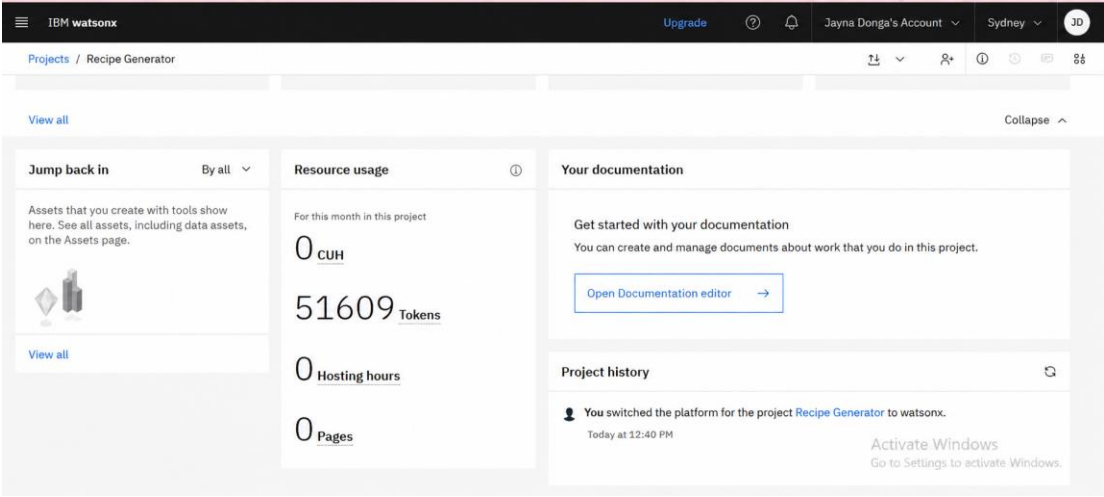
Token Usage

11,761 token are used

Before Prompt



After the Prompt



Novelty and Uniqueness

The proposed **Document Q&A RAG Agent** combines semantic search, Retrieval-Augmented Generation (RAG), and AI-powered recipe personalization into a single intelligent cooking assistant.

Recipe-Aware RAG – Retrieves relevant recipes from unstructured documents instead of relying only on the LLM's knowledge.

Personalized Recipe Generation – Adapts recipes based on dietary restrictions, available ingredients, cooking time, and cuisine preferences.

Conversational Cooking Assistant – Allows users to interact naturally and receive context-aware recipe recommendations.

Smart Recipe Adaptation – Suggests ingredient substitutions, step-by-step cooking instructions, and nutritional insights.

Scalable Knowledge Base – Supports continuous ingestion of cookbooks, blogs, PDFs, and personal recipe collections without retraining the model.

Enterprise-Ready – Built using **LangFlow**, **IBM watsonx.ai**, and **Granite Models**, ensuring scalability, reliability, and efficient AI-powered retrieval.

This makes the solution an intelligent recipe assistant that goes beyond keyword search by delivering personalized and context-aware cooking guidance.

Git Hub Link

· Please create public github repository (Enable Add README button while creating repository)

with format –

Please test and Paste the working Github URL –

<https://github.com/jaynambict-maker/IBM-SkillsBuild-Project/>

Make sure you have Uploaded following three files into your github repository

1) app.json file , 2) yourproblemstatement.pdf file 3) your projectpresntation.pptx file 4)any other files

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Future Scope

1.Integration with Wearables & Health Devices –

The system can be integrated with smartwatches and fitness trackers to collect real-time data such as steps, calories burned, heart rate, and sleep patterns. This will enable the Nutrition Agent to provide more accurate and dynamic diet recommendations based on actual user activity and health metrics.

2.AI-Powered Voice Assistant & Multilingual Support –

Enhancing the system with a voice-enabled assistant and support for multiple languages will improve accessibility. Users can interact naturally in their preferred language, making the solution more inclusive and suitable for diverse populations, especially in regions with varied linguistic backgrounds.

Certificates Earned:

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IBM SkillsBuild	Completion Certificate
	This certificate is presented to jayna donga for the completion of Lab: Troubleshoot Your Code Using IBM Bob (ALM-COURSE_4071307) According to the Adobe Learning Manager system of record
Completion date: 23 Jun 2026 (GMT)	Learning hours: 30 mins

Thank You!

Thank you for your time and interest.